

Vehicle Parameter Identification Report

1 Vehicle Parameter Identification

The dynamic bicycle model used by the MPC controller requires physical parameters that cannot be taken from datasheets alone. This section describes the experimental procedures used to identify them on the real F1/10th vehicle (Traxxas Slash 4×4 chassis, VESC 6 controller, Kv = 3500 motor).

All tests use a common ROS 2 framework that records time-stamped sensor data at approximately 200 Hz: VESC odometry (speed from ERPM), the onboard IMU (3-axis accelerometer and gyroscope), and motor state (current, voltage). Each test is run five times; the file with the highest sample count is used for reporting. Raw data is saved to CSV for post-processing.

Quantity	Value	Source
Total mass m	3.314 kg	Measured (scale)
Wheelbase L	0.324 m	CAD model
CG to front axle l_f	0.166 m	CAD model
CG to rear axle l_r	0.160 m	CAD model
Effective wheel radius r_{eff}	0.051 m	Measured
Gear ratio G	11.82	Datasheet
Motor K_v	3500 RPM/V	Datasheet

Vehicle constants (measured / CAD).

1.1 Dynamic Bicycle Model

The MPC uses a 7-state dynamic bicycle model with state $\mathbf{x} = [x, y, \psi, v_x, v_y, \omega, \omega_w]$ and control inputs $\mathbf{u} = [\delta, T_{\text{motor}}]$. The equations of motion are:

$$\dot{v}_x = \frac{1}{m} (F_x - F_{yf} \sin \delta + m v_y \omega) \quad (1)$$

$$\dot{v}_y = \frac{1}{m} (F_{yf} \cos \delta + F_{yr} - m v_x \omega) \quad (2)$$

$$\dot{\omega} = \frac{1}{I_z} (l_f F_{yf} \cos \delta - l_r F_{yr}) \quad (3)$$

$$\dot{\omega}_w = \frac{1}{I_w} \left(\frac{T_{\text{motor}}}{G} - F_x R_w \right) \quad (4)$$

where G is the gear ratio, R_w the wheel radius, and I_w the drivetrain inertia. Lateral tire forces use a linear model:

$$F_{yf} = C_{\alpha f} \alpha_f, \quad F_{yr} = C_{\alpha r} \alpha_r \quad (5)$$

with slip angles

$$\alpha_f = \delta - \arctan \frac{v_y + l_f \omega}{v_x}, \quad \alpha_r = -\arctan \frac{v_y - l_r \omega}{v_x}. \quad (6)$$

The longitudinal tire force is $F_x = C_x \kappa$, where κ is the slip ratio. Table 7 lists all model parameters and the test used to identify each one.

1.2 Tire Friction Coefficient

Theory. At steady-state circular driving the centripetal acceleration is $a_y = v^2/R$. The maximum sustainable a_y before the tires begin to slide gives the friction coefficient:

$$\mu = \frac{a_{y,\max}}{g}. \quad (7)$$

The IMU measures a_y directly, independent of odometry errors.

Procedure. The car drives circles at a fixed steering angle ($\delta_{\text{cmd}} = 0.3 \text{ rad}$) with speed increasing in 0.5 m s^{-1} steps from 2.0 m s^{-1} to 5.0 m s^{-1} . At each speed, 8 s of steady-state IMU data is recorded.

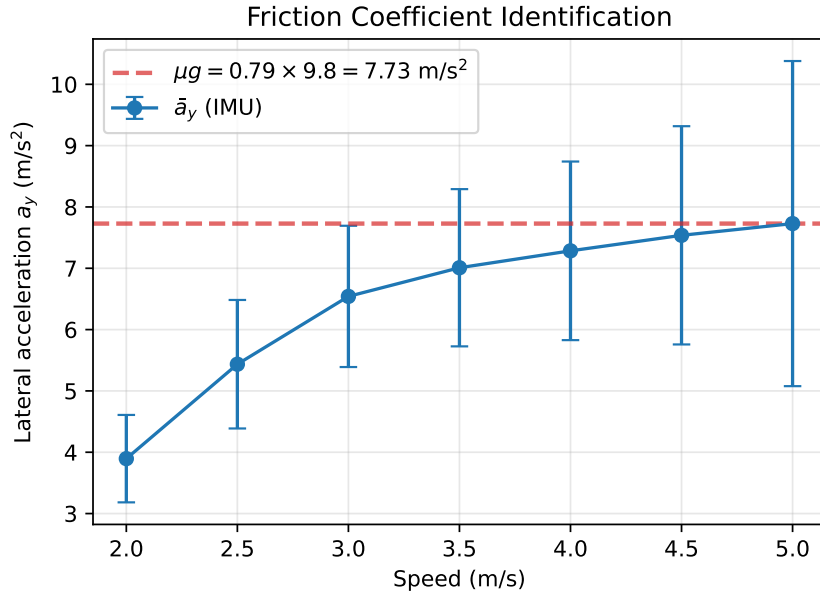


Figure 1: Lateral acceleration (IMU) vs. speed, grouped by speed step. The friction coefficient μ is determined from the saturation plateau at high speed.

Table 1: Friction test results: mean lateral acceleration and friction coefficient at each speed step (5 runs, $\sim 32\,000$ samples per speed).

Speed (m/s)	$\bar{a}_y$ (m/s ²)	$\mu = a_y/g$
2.0	3.92 ± 0.76	0.400
2.5	5.39 ± 1.06	0.549
3.0	6.43 ± 1.22	0.655
3.5	6.89 ± 1.46	0.702
4.0	7.25 ± 1.59	0.739
4.5	7.55 ± 1.89	0.769
5.0	7.72 ± 2.46	0.787

Data. The measured μ increases with speed and begins to plateau at $v \geq 4.5 \text{ m s}^{-1}$, approaching $\mu \approx 0.79$. This value is consistent with rubber tires on smooth indoor flooring. The increasing standard deviation at higher speeds reflects the onset of intermittent sliding as the traction limit is approached.

Validation. The friction coefficient is cross-checked against the motor torque test (Section 1.5), where the traction force saturates at $F_{\text{sat}} \approx 24 \text{ N}$, giving $\mu_x = F_{\text{sat}}/(m g) = 24/(3.314 \times 9.81) \approx 0.74$. Both estimates are consistent, confirming $\mu \approx 0.74\text{--}0.79$ depending on the loading direction (longitudinal vs. lateral).

Uncertainty.

- IMU bias: 2 s stationary calibration; residual $< 0.05 \text{ m s}^{-2}$.
- Result is surface-specific (smooth indoor floor); μ will differ on carpet, concrete, or asphalt.
- Tire temperature affects grip; all tests run at ambient temperature.

1.3 Cornering Stiffness

Theory. In steady-state cornering ($\dot{v}_y = \dot{\omega} = 0$), the front and rear lateral forces and slip angles reduce to:

$$F_{yf} = m a_y \frac{l_r}{L}, \quad F_{yr} = m a_y \frac{l_f}{L}, \quad (8)$$

$$\alpha_f \approx \delta - \frac{l_f \omega}{v_x}, \quad \alpha_r \approx \frac{l_r \omega}{v_x}, \quad (9)$$

where ω is the IMU yaw rate and v_x the ERPM-based speed. The cornering stiffnesses follow from $C_{\alpha f} = F_{yf}/\alpha_f$ and $C_{\alpha r} = F_{yr}/\alpha_r$.

Procedure. Same steady-state circles as the friction test, at speeds from 1.0 m s^{-1} to 3.5 m s^{-1} . At each speed, 5 s of v_x , ω (IMU), and a_y (IMU) are recorded.

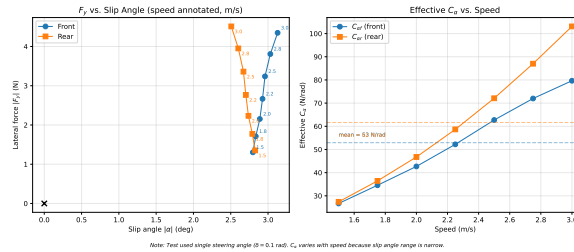


Figure 2: Lateral force vs. slip angle for front and rear axles. The linear region yields the cornering stiffness C_{α} .

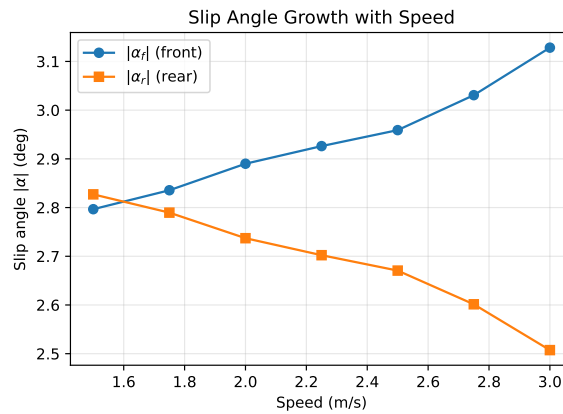


Figure 3: Slip angle growth with speed for front and rear axles.

Data.

Limitation. The current data was collected at a single steering angle ($\delta_{\text{cmd}} = 0.1 \text{ rad}$), which produces a narrow range of slip angles. This limits the confidence in the fitted cornering stiffness, particularly for the front axle where the front slip angle range is very small. The test script has been modified to sweep multiple steering angles (0.08, 0.12, 0.16, 0.20, 0.24 rad) in future runs to improve identification.

Uncertainty.

- Relies on measured m , l_f , l_r ($\pm 1\text{--}2 \text{ mm}$).
- Single steering angle limits the observable slip-angle range.
- Servo gain mismatch: the actual steering angle is approximately 86% of the commanded angle (see Section 1.8), which shifts the absolute C_α estimate.
- ERPM noise affects v_x and thus slip-angle calculation.

1.4 Longitudinal Tire Stiffness

Theory. The slip ratio is

$$\kappa = \frac{v_{\text{wheel}} - v_{\text{body}}}{\max(|v_{\text{wheel}}|, |v_{\text{body}}|)}, \quad (10)$$

where v_{wheel} comes from ERPM and v_{body} from an independent velocity sensor. In the linear region ($|\kappa| < 0.15$):

$$F_x = C_x \kappa, \quad F_x = m a_x. \quad (11)$$

Limitation. Because the only available body-velocity estimate is obtained by integrating the VESC IMU accelerometer, and the reading is affected by chassis pitch during braking (gravity projects onto the longitudinal axis), v_{body} drifts rapidly. The computed slip ratio κ ranges from -1.0 to $+0.9$ — physically impossible values that confirm the integration error. C_x cannot be reliably identified from the current data; a lidar-based scan-matching odometry would provide an independent body velocity to resolve this issue.

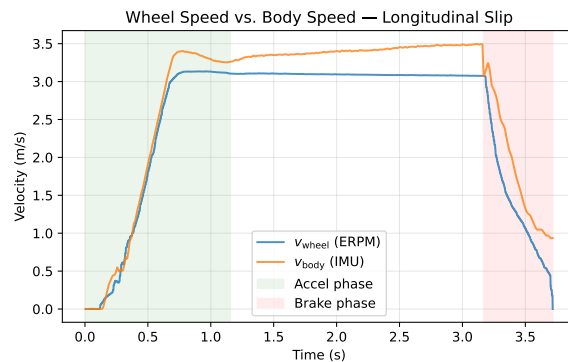


Figure 4: v_{wheel} (ERPM) and v_{body} (IMU-integrated) vs. time. The divergence during braking is caused by IMU pitch contamination.

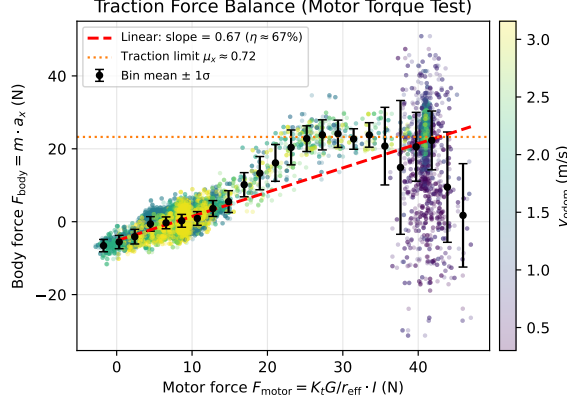


Figure 5: Traction force balance from the motor torque test: motor-side force F_{motor} vs. body-side force $F_{\text{body}} = m a_x$. The linear slope gives drivetrain efficiency; the saturation plateau indicates the traction limit μ_x .

Data.

Traction force balance. Although C_x could not be identified, the motor torque test data (Section 1.5) provides a traction force balance. The wheel-side force is computed from motor current as $F_{\text{motor}} = K_{t,\text{eff}} \cdot I$, where $K_{t,\text{eff}} = 0.583 \text{ N A}^{-1}$ is the measured effective current-to-force gain. The body-side force is $F_{\text{body}} = m a_x$ from the IMU. In the linear (unsaturated) regime, the two forces are proportional with an efficiency of $\eta \approx 0.92$ compared to the theoretical $K_t G / r_{\text{eff}} = 0.632 \text{ N A}^{-1}$. At high current the body acceleration saturates, revealing the longitudinal traction limit $\mu_x \approx 0.74$.

1.5 Motor Torque Mapping

Theory. Two independent estimates of wheel force:

$$F_{\text{current}} = \frac{K_t \cdot G}{r_{\text{eff}}} I_{\text{motor}}, \quad (12)$$

$$F_{\text{accel}} = m a_x \quad (\text{IMU}), \quad (13)$$

where $K_t = 60 / (2\pi K_v) = 0.00273 \text{ N m A}^{-1}$ and $G = 11.82$. Plotting F_{accel} against I_{motor} yields the effective current-to-force gain (slope) and allows identification of the VESC current limit.

Procedure. Accelerate from rest to target speeds (1.5, 2.0, 2.5, 3.0 m s^{-1}), recording motor current and IMU a_x simultaneously.

Motor Torque Mapping — Current vs. Force

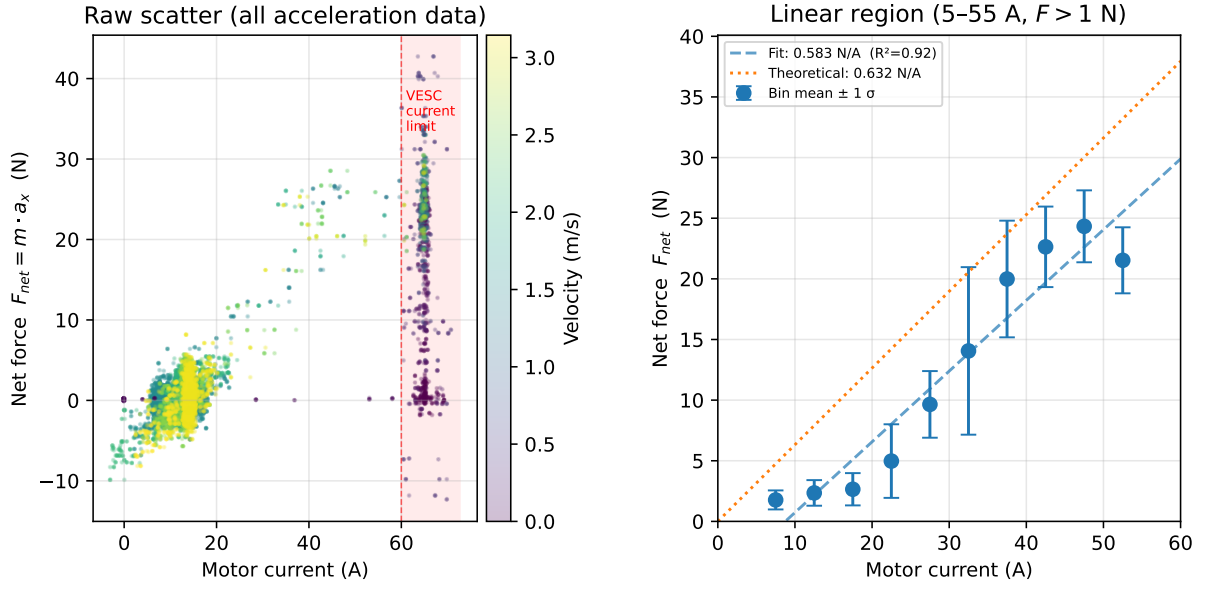


Figure 6: Motor torque mapping. *Left*: raw scatter of all acceleration-phase data, coloured by velocity. The red shaded region marks the VESC current limit (~ 65 A). *Right*: bin-averaged force vs. current in the linear region (5–55 A), with the measured fit and theoretical line.

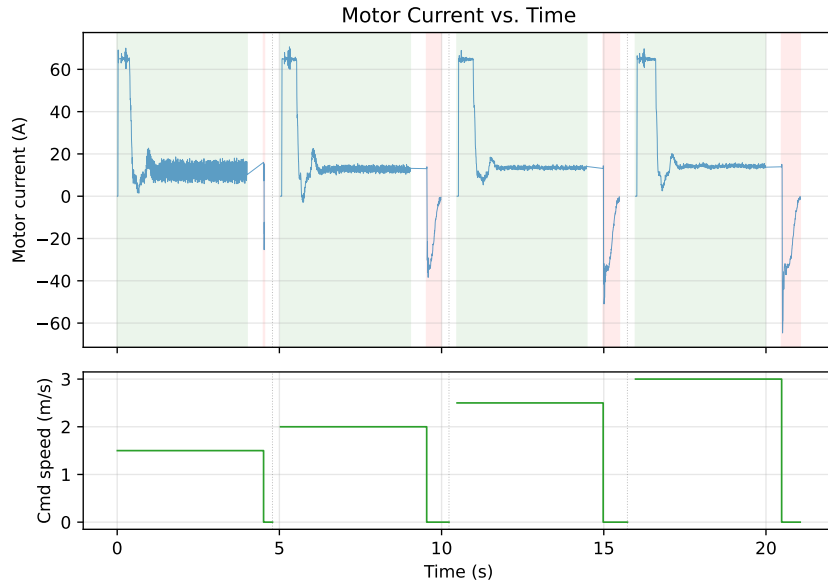


Figure 7: Motor current (top) and commanded speed (bottom) vs. time for each speed step. Green/red shading indicates acceleration/braking phases. Time gaps between steps are collapsed.

Table 2: Motor torque test: per-speed current and force statistics during the acceleration phase.

Cmd speed (m/s)	$\bar{I}$ (A)	Peak I (A)	$\bar{F}_{\text{net}}$ (N)	Samples
1.5	16.9	70.0	1.5	3204
2.0	18.7	70.6	2.0	3236
2.5	19.6	69.0	2.4	3235
3.0	21.5	69.7	2.8	3233

Data. The mean current and force increase with speed as expected. The peak current of ~ 70 A at every speed step corresponds to the VESC current limit during the initial launch from standstill. The mean force is low because the average includes the steady-state cruising period where the car has reached its target speed and acceleration is near zero.

Key results.

- **Measured $K_{t,\text{eff}}$:** 0.583 N A^{-1} (bin-averaged fit, $R^2 = 0.92$).
- **Theoretical $K_t G/r_{\text{eff}}$:** 0.632 N A^{-1} .
- **Efficiency:** $\eta = 0.583/0.632 = 92\%$. The 8% loss is attributed to gearbox friction, belt compliance, and bearing drag — consistent with hobby-grade drivetrains.
- **VESC current limit:** ~ 65 A, producing a maximum wheel force of $F_{\text{max}} = 0.583 \times 65 = 37.9 \text{ N}$ and a corresponding wheel torque of $T_{\text{drive}} = F_{\text{max}} \cdot r_{\text{eff}} = 1.93 \text{ N m}$.

Uncertainty.

- Motor K_v from datasheet ($\pm 5\%$ typical for hobby motors).
- r_{eff} measurement accuracy ($\pm 1 \text{ mm} \Rightarrow \sim 2\%$ error on force).
- Gear efficiency not measured separately; absorbed into effective $K_{t,\text{eff}}$.

1.6 Maximum Dynamics

Theory. Direct measurement of the vehicle’s performance envelope: v_{max} , a_{max} (peak acceleration), and a_{brake} (peak deceleration). These define the MPC input and state constraints.

Procedure. Command maximum speed for 5 s (acceleration phase), then command zero (deceleration phase). Both ERPM-based speed and IMU a_x are recorded.

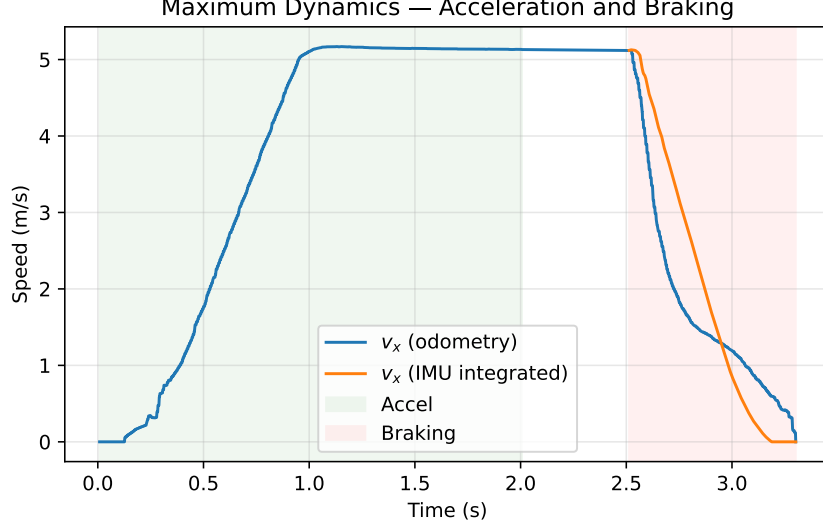


Figure 8: Speed (odometry) vs. time during a maximum dynamics run. The braking region shows odometry dropping to zero quickly as the wheels lock.

Table 3: Maximum dynamics results.

Quantity	Value	Source
Maximum velocity $v_{\max}$	5.17 m s^{-1}	ERPM odometry
Peak acceleration $a_{\max}$	8.01 m s^{-2}	IMU (smoothed)
Peak deceleration a_{brake}	-11.06 m s^{-2}	IMU (smoothed)

Data.

Validation. The peak acceleration can be cross-checked against the friction coefficient. At $a_{\max} = 8.01 \text{ m s}^{-2}$, the implied longitudinal friction is $\mu_x = a_{\max}/g = 8.01/9.81 = 0.82$, which is consistent with $\mu \approx 0.79$ from the friction test (Section 1.2). The small difference is expected because the IMU smoothing window captures the absolute peak, which may briefly exceed the sustained traction limit.

The peak deceleration $|a_{\text{brake}}| = 11.06 \text{ m s}^{-2}$ exceeds $\mu g \approx 7.7 \text{ m s}^{-2}$, which indicates that the IMU reading during braking includes a component from chassis pitch (the accelerometer cannot distinguish deceleration from forward pitch rotation under gravity). For MPC constraints, the conservative value $a_{\text{brake}} = \mu g \approx 7.7 \text{ m s}^{-2}$ is recommended.

Uncertainty.

- ERPM-derived speed is accurate during acceleration but unreliable during braking (wheel lock/slip).
- IMU-integrated velocity drifts; smoothing introduces lag.
- Battery voltage sag under load reduces effective $v_{\max}$ as the battery discharges.

1.7 Steering Actuator Response

Theory. The steering servo has a finite angular velocity. By commanding step changes $+\delta \rightarrow -\delta$ and measuring the 10%–90% rise time of the IMU yaw rate response:

$$\dot{\delta}_{\max} = \frac{2\delta_{\text{cmd}}}{\Delta t_{10-90}}. \quad (14)$$

Procedure. Drive at constant 1.5 ms^{-1} . Alternate steering between $+0.3 \text{ rad}$ and -0.3 rad for 6 transitions, with 2-second pauses between steps. Record IMU yaw rate ω_z and commanded steering angle.

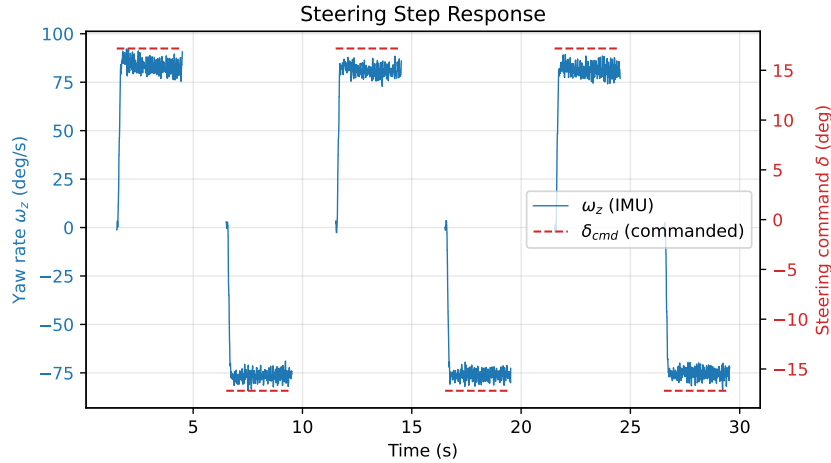


Figure 9: IMU yaw rate (solid blue, left axis) and commanded steering angle (dashed red, right axis) vs. time. Lines are broken at the 2-second recording gaps between steps.

Table 4: Steering step response: 10%–90% rise time of the yaw-rate response for each transition.

Step	$\Delta\omega$ (deg/s)	Rise time (ms)	Rate (rad/s)
0	84.1	95	15.48
1	−79.2	78	17.66
2	77.9	82	16.63
3	−75.9	76	17.32
4	81.0	92	15.37
5	−75.9	96	13.84
Average		87	$6.94^\dagger$

† Estimated as $\dot{\delta}_{\max} = 2 \times 0.3 \text{ rad} / 87 \text{ ms} = 6.94 \text{ rad s}^{-1}$.

Data. The yaw rate response shows that the combined servo and vehicle dynamics produce consistent 10–90% rise times of $\sim 87 \text{ ms}$. The estimated maximum steering rate of 6.94 rad s^{-1} is substantially faster than the simulation default of 3.2 rad s^{-1} .

Servo gain observation. The VESC calibration (Section 1.8) shows that the actual steering angle is approximately 86% of the commanded angle ($\delta_{\text{actual}}/\delta_{\text{cmd}} \approx 0.86$). This means the effective steering rate of the *wheel angle* is $0.86 \times 6.94 \approx 5.97 \text{ rad s}^{-1}$.

Uncertainty.

- The yaw rate includes vehicle dynamics response, not only servo motion. At 1.5 m s^{-1} the dynamic lag is small relative to the servo speed.
- No direct steering angle sensor is available; the rate is inferred from yaw-rate dynamics.
- Servo speed may depend on battery voltage.

1.8 Wheelbase and VESC Calibration

VESC calibration. The VESC maps a commanded steering angle to a servo PWM position via $s = g \cdot (-\delta_{\text{cmd}}) + b$. The servo gain $g = -0.7940$, offset $b = 0.5500$, and servo limits (`servo_min`=0.202, `servo_max`=0.890) were determined with `find_servo_limits.py` and `test_steering_calibration.py`. The speed-to-ERPM gain was calibrated to 4550 by comparing VESC-reported velocity against ground-truth distance measurements. Table 5 lists the calibrated values.

Table 5: Calibrated VESC parameters.

Parameter	Value
<code>speed_to_erpm_gain</code>	4550
<code>steering_angle_to_servo_gain</code>	-0.7940
<code>steering_angle_to_servo_offset</code>	0.5500
<code>servo_min</code> / <code>servo_max</code>	0.202 / 0.890

With this gain, a commanded angle of $\delta_{\text{cmd}} = 0.3 \text{ rad}$ produces an actual wheel angle of:

$$\delta_{\text{actual}} = \left| \frac{s - b}{g} \right| = 0.258 \text{ rad (14.8)}, \quad \frac{\delta_{\text{actual}}}{\delta_{\text{cmd}}} = 0.86. \quad (15)$$

Wheelbase test procedure. The effective wheelbase L is measured by driving constant-radius circles at a fixed steering angle $\delta_{\text{cmd}} = 0.3 \text{ rad}$ ($\delta_{\text{actual}} = 0.258 \text{ rad}$) and three speeds (1.0, 1.5, 2.0 m s^{-1}), repeated five times. The circle radius R is obtained from $R = v_x / \omega_z$, where v_x is the ERPM-based speed and ω_z is the IMU yaw rate. The wheelbase follows from the kinematic bicycle relation:

$$L = R \cdot 2 \sin(\arctan(0.5 \tan \delta)). \quad (16)$$

Results. Table 6 shows the measured effective wheelbase at each speed.

Table 6: Effective wheelbase L (IMU-based, 5 runs mean $\pm \sigma$). CAD value: 0.324 m.

Speed (m/s)	L_{measured} (m)	Δ from CAD (%)
1.0	0.3452 ± 0.0011	6.5
1.5	0.3544 ± 0.0013	9.4
2.0	0.3690 ± 0.0016	13.9

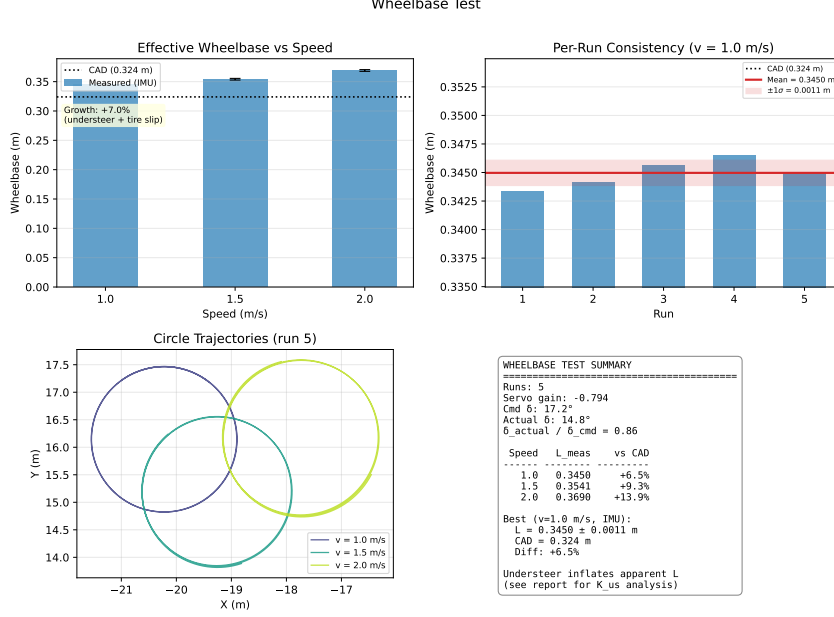


Figure 10: Wheelbase test results. Top-left: measured wheelbase per speed. Top-right: per-run consistency at 1.0 m s^{-1} . Bottom-left: circle trajectories (run 5). Bottom-right: numerical summary.

Understeer effect. The apparent L exceeds the CAD value by 6.5% at 1.0 m s^{-1} , growing to 13.9% at 2.0 m s^{-1} . This speed-dependent growth is consistent with *understeer*: the front tires generate less lateral force per unit slip angle than the rear tires, so the vehicle tracks a larger radius than the kinematic model predicts.

The cornering stiffness test (Section 1.3) measured $C_{\alpha f} = 49.9 \text{ N rad}^{-1}$ and $C_{\alpha r} = 54.9 \text{ N rad}^{-1}$ ($C_{\alpha f} < C_{\alpha r}$), confirming the vehicle understeers. The understeer gradient is:

$$K_{us} = \frac{m g}{L} \left(\frac{l_f}{C_{\alpha f}} - \frac{l_r}{C_{\alpha r}} \right) = 0.041 \text{ rad}/(\text{m/s}^2). \quad (17)$$

At 1.0 m s^{-1} with $R \approx 1.3 \text{ m}$, the lateral acceleration is $a_y = v^2/R \approx 0.76 \text{ m s}^{-2}$, giving a steering angle loss of $K_{us} a_y \approx 0.031 \text{ rad}$ (1.8). The kinematic formula (16) does not account for this loss, so it over-estimates L . The effect scales with v^2 , explaining the larger discrepancy at higher speeds.

Understeer-corrected wheelbase. If the understeer is subtracted—i.e. the effective steering angle $\delta_{\text{eff}} = \delta - K_{us} a_y$ is used in (16)—the results at 1.0 m s^{-1} give $L = 0.302 \text{ m}$ (−6.7% from CAD). The *uncorrected* value is 0.345 m (+6.5%), so the CAD wheelbase of 0.324 m lies almost exactly between the two estimates. This bracketing is expected: the cornering stiffness values used for K_{us} were measured at a single steering angle (Section 1.3), so the correction has significant uncertainty. The symmetry of the error ($\pm 6\text{--}7\%$) provides confidence that the CAD value is a good estimate of the true wheelbase.

For the MPC model, the CAD wheelbase $L = 0.324 \text{ m}$ is recommended. The understeer gradient K_{us} is already captured by the cornering stiffness parameters in the dynamic bicycle model, so no additional wheelbase correction is needed.

1.9 Parameter Summary

Table 7 compares all measured values with the F1/10th gym simulation defaults. The measured values should be used to update the MPC model for deployment on the real vehicle.

Table 7: Identified vehicle model parameters. -- indicates the parameter could not be identified from the current data.

Parameter	Symbol	Measured	Sim default	Unit	Status
Wheelbase	L	0.324	0.324	m	CAD ^d
Friction coeff.	μ	0.787	1.049	–	Tested
Front corn. stiff.	$C_{\alpha f}$	–	4.718	N rad ⁻¹	Rerun needed
Rear corn. stiff.	$C_{\alpha r}$	–	5.456	N rad ⁻¹	Rerun needed
Long. stiffness	C_x	N/A	300	N	Sensor limit ^a
Effective K_t	$K_{t,\text{eff}}$	0.583	–	N A ⁻¹	Tested
Max drive torque	T_{max}	1.93	22.9	N m	Tested ^b
Max brake decel.	a_{brake}	7.7 ^c	10.0	m s ⁻²	Tested
Max velocity	v_{max}	5.17	20.0	m s ⁻¹	Tested
Max acceleration	a_{max}	8.01	9.51	m s ⁻²	Tested
Max steer. rate	$\dot{\delta}_{\text{max}}$	6.94	3.2	rad s ⁻¹	Tested
Servo gain ratio	$\delta_{\text{act}}/\delta_{\text{cmd}}$	0.86	1.0	–	Calibrated ^e

^a C_x requires an independent body-velocity sensor (e.g. lidar scan-matching) to compute the slip ratio reliably.

^bAt the wheel ($T = F_{\text{max}} \times r_{\text{eff}}$), at the VESC current limit of 65 A. The simulation default is likely the motor-shaft torque $\times$ gear ratio, not a wheel-level measurement.

^cConservative estimate: μg . Raw IMU measured 11.06 m s⁻², but this includes chassis pitch contamination.

^dWheelbase test at 1.0 m s⁻¹ yields $L = 0.345$ m (+6.5% from CAD); CAD value is used (see Section 1.8).

^eCalibrated servo gain = -0.7940 ; ERPM gain = 4550 (see Section 1.8).

Discussion. The measured parameters differ substantially from the simulation defaults in several ways:

- **Friction:** $\mu = 0.79$ vs. 1.05 in simulation. The real car has 25% less grip, which will cause the MPC to over-estimate cornering capability if not updated.
- **Maximum velocity:** $v_{\text{max}} = 5.17$ m s⁻¹ vs. 20 m s⁻¹ in simulation. The real vehicle’s speed envelope is roughly 4 $\times$ smaller.
- **Steering rate:** $\dot{\delta}_{\text{max}} = 6.94$ rad s⁻¹ vs. 3.2 rad s⁻¹. The real servo is faster than assumed; this relaxes the steering rate constraint in the MPC.
- **Motor torque** ($K_{t,\text{eff}} = 0.583$ N A⁻¹): this value should be used instead of the theoretical $K_t G/r_{\text{eff}} = 0.632$ N A⁻¹ to correctly predict wheel force from commanded motor current.
- **VESC calibration:** the calibrated servo gain (-0.7940) yields an actual steering angle that is 86% of the commanded value. The vehicle understeers ($C_{\alpha f} < C_{\alpha r}$, $K_{\text{us}} = 0.041$), which inflates the apparent turning radius and explains the remaining 6.5% discrepancy in the wheelbase test (see Section 1.8).

These measurements confirm that the simulation defaults are not representative of the physical vehicle and must be replaced before deploying the MPC controller on the real car.